Your universe at the best fit: thick ordinary-matter shell plus a CoE core (relativistic fit)

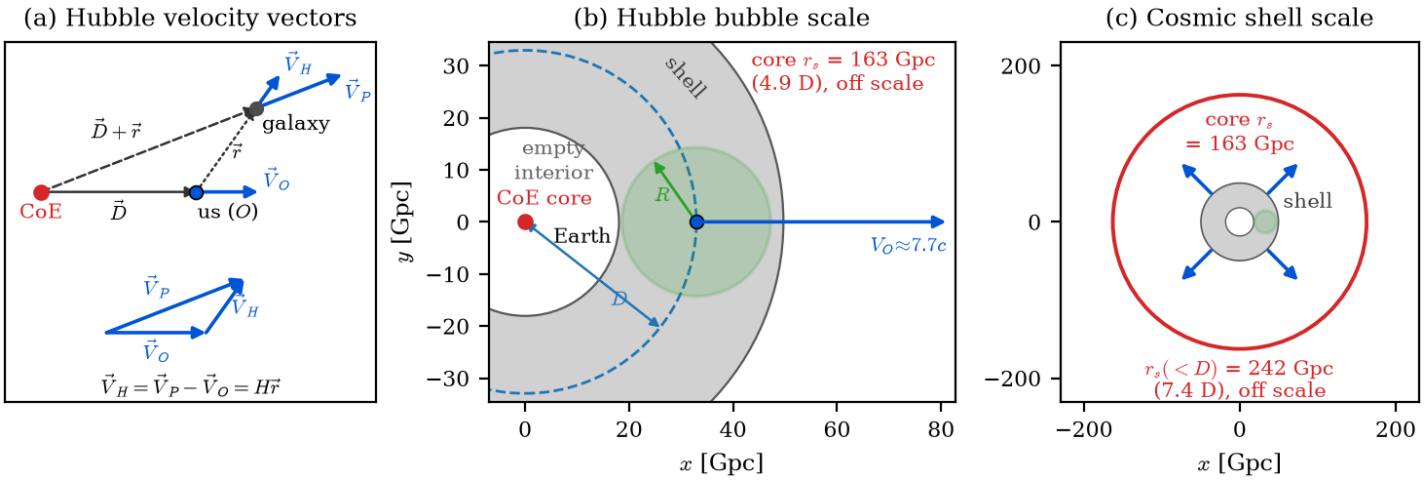


Figure 1: Three scales at the relativistic best fit; (b) and (c) are to scale in Gpc. (a) Hubble velocity vectors. In a Sedov–Taylor blast wave (Taylor 1950; Sedov 1959) $\vec{V} = k \times$(position from the CoE), so we move with $\vec{V}_O = k\vec{D}$ and a galaxy with $\vec{V}_P = k(\vec{D}+\vec{r})$; then $\vec{V}_P - \vec{V}_O = k\vec{r}$ is Hubble's law, so $k = H$ and the CoE drops out. (b) Hubble bubble: ordinary-matter shell (grey, $\Omega_s = 0.049$) from $R_{\rm in} = 18.1$ to $R_{\rm out} = 49.7$ Gpc around an empty interior and the CoE core; dashed circle $D = 32.9$ Gpc through Earth; green vision sphere $R = 14.3$ Gpc; Earth moves along $+x$ (anti-CoE, $\ell_0 = 13.5°$) at $V_O = H_0 D/c \approx 7.7\,c$. (c) Cosmic shell: the core's Schwarzschild radius $r_s = 163$ Gpc ($4.9\,D$); r_s of all mass inside D is 242 Gpc ($7.4\,D$), off scale. Blue arrows are velocities. The fit is exact relativistic (LTB), with ordinary matter only.

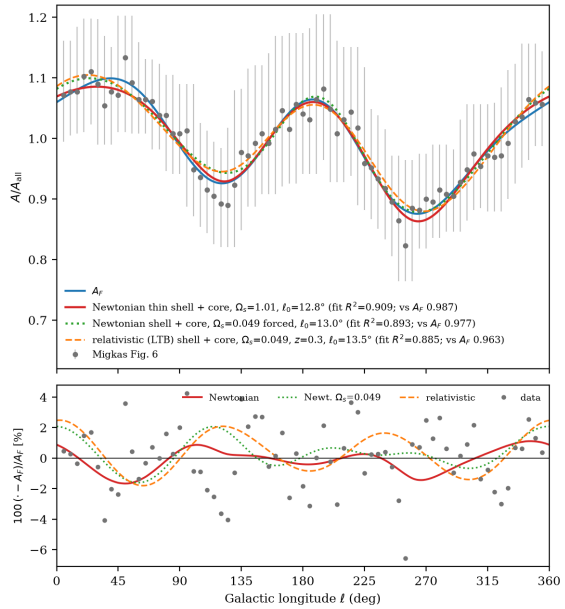


Figure 2: Migkas points with error bars, A_F and model A, normalised to the sky mean. Solid red: Newtonian thin shell with core ($\Omega_s = 1.01$, $R^2 = 0.909$). Dotted green: Newtonian with $\Omega_s = 0.049$ forced, $R^2 = 0.893$ (0.54 geometry only). Dashed orange: relativistic (LTB) ordinary-matter shell with core ($\Omega_s = 0.049$, all parameters free), sampled at the Migkas cluster distance ($z = 0.3$, 1.19 Gpc), $R^2 = 0.885$ (0.57 without the sine). Lower panel: percent differences from A_F.

Table 1: Best-fit values in your sheet's cells. Newtonian: integrated model (G real, from $0.01\,t_0$, fixed in space, $H_0 = 70$), thin shell $D-R$ to $D+R$. Newtonian Ω_s=0.049: same model with the shell density forced to ordinary matter. Relativistic: exact LTB, ordinary-matter shell; D, M_c, $R_{\rm in}$, $R_{\rm out}$, ℓ_0, Amp, φ all free (shell $\geq R$ each side of Earth); $H = 78/66$ km s^{-1} Mpc^{-1} along/across the axis. R^2 in brackets: geometry only (no sine). B27 is all mass inside D.

Cell	Quantity	Newtonian, Ω_s free	Newtonian, Ω_s=0.049	Relativistic, Ω_s=0.049, M_c free
C23	D/R	6.554	3.439	2.303
D23	L_c/D (core size)	0.0050 (bound)	0.0050	0.0050
B29	R	4.4127×10^{23} km (14.30 Gpc)	same	same
B28	D	2.8919×10^{24} km (93.72 Gpc)	1.5174×10^{24} km (49.17 Gpc)	1.0164×10^{24} km (32.94 Gpc)
B30	L_c	1.45×10^{22} km (0.469 Gpc)	7.59×10^{21} km (0.246 Gpc)	5.08×10^{21} km (0.165 Gpc)
B27	$M(< D)$, total	3.72×10^{56} kg	4.56×10^{54} kg	5.04×10^{54} kg
		$1.87 \times 10^{26}\ M_\odot$	$2.29 \times 10^{24}\ M_\odot$	$2.53 \times 10^{24}\ M_\odot$
	M_c, core	2.47×10^{54} kg	2.31×10^{53} kg	3.38×10^{54} kg
		$1.24 \times 10^{24}\ M_\odot$	$1.16 \times 10^{23}\ M_\odot$	$1.70 \times 10^{24}\ M_\odot$
	$M_c/M_{\rm crit}(D)$	0.00265	0.00171	0.083
	$M_{\rm shell}(< D)$	3.70×10^{56} kg	4.33×10^{54} kg	1.66×10^{54} kg
	$M_{\rm shell}$, total	8.71×10^{56} kg	1.70×10^{55} kg	6.48×10^{54} kg
	M within R	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg
	$M_{\rm hole}$	5.74×10^{56} kg	2.28×10^{54} kg	3.27×10^{53} kg
BA57	Amp	−0.1240	−0.0622	−0.0558
BA59	φ	1.264 rad	0.929 rad	0.848 rad
	ℓ_0	12.82°	13.0°	13.5°
	Ω_s	1.012	0.049	0.049
	ρ_s	9.32×10^{-27} kg m^{-3}	4.51×10^{-28} kg m^{-3}	4.51×10^{-28} kg m^{-3}
	$R_{\rm in}$	79.4 Gpc ($D-R$)	34.5 Gpc	18.1 Gpc
	$R_{\rm out}$	108.0 Gpc ($D+R$)	70.3 Gpc	49.7 Gpc
	r_s, core	119 Gpc (1.27 D)	11 Gpc (0.2 D)	163 Gpc (4.9 D)
	r_s, all mass $< D$	17,907 Gpc (191 D)	219 Gpc (4.5 D)	242 Gpc (7.4 D)
	$(V_O/v_{\rm esc})^2$ / $2E/c^2$	0.901 (bound)	8.0 (unbound)	+45 (unbound)
	Turnaround radius	991 Gpc (10.6 D)	none	none
	Age t_0	13.8 Gyr	13.9 Gyr	13.0 Gyr
	δ across sphere	0	0	0
	fit R^2 vs Migkas	0.909	0.893 (0.54)	0.885 (0.57)
	fit R^2 vs A_F	0.987	0.977	0.963

References. Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159. Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (New York: Academic Press).